

Village Clean-Heating *Transition Sandbox*

Coal-to-X Household-Level Clean Heating Transition Simulation Sandbox

RESEARCH QUESTION • HOOK & GAP

Less pollution, but still unaffordable — which path?

A PUBLIC DEBATE

Is clean heating good?

Regional stats dominate; household income, housing, and heating costs differ.

B THIS STUDY

Which coal-to-X route fits this household?

Interactive comparison of gas / heat-pump routes, with compliance and subsidy shocks.



POLICY SUCCESS CAN COEXIST WITH HOUSEHOLD AFFORDABILITY PRESSURE.

SCOPE

Converted and unconverted clean-heating farmers in North China; students and other interested public users.

HYPOTHESES

H1 Farmers can find a coal-to-X path that fits their own household. H2 Students and other users improve their understanding after using the sandbox.

RESEARCH PROGRESS · GAP · CONTRIBUTION

Evidence is strong.
Decision support is still thin.

RESEARCH PROGRESS

Previous studies show clear pollution and carbon benefits, while policy simulation can connect household behavior with larger outcomes.

Liu et al. · Yuan et al. · Wang

LIMITATIONS

Affordability, infrastructure, fiscal pressure, price risk and coal rebound are recognized, but most work remains regional or policy-level.

Few tools compare gas, electricity, heat pumps and insulation from household inputs.

MY CONTRIBUTION

I turn the policy problem into an interactive village-level sandbox using income, house size, heating cost and affordability pressure.

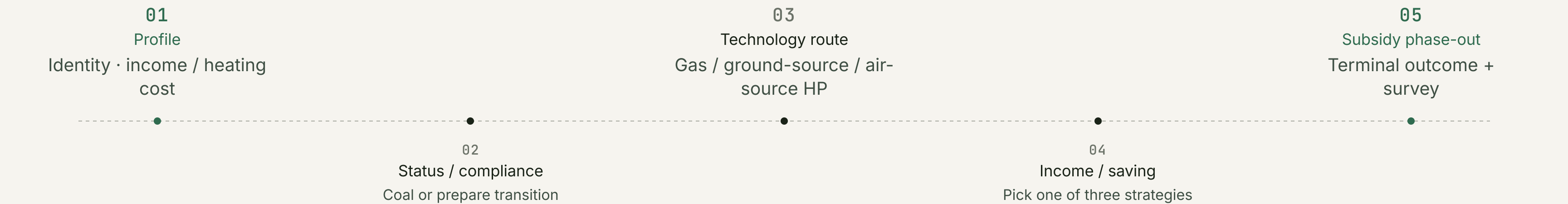
The question shifts from "Is clean heating good?" to "Which coal-to-X path fits this village?"



METHOD • ARTIFACT + DESIGN

Five-turn simulation + pre/post surveys + AI analysis

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SAMPLE TARGET

20+ participants

Students + rural users + public URL

IDENTITY GROUPS

4 types

Student / Converted / Unconverted / Other

DATA STORAGE

Supabase

Anonymous UUID · surveys + AI analysis

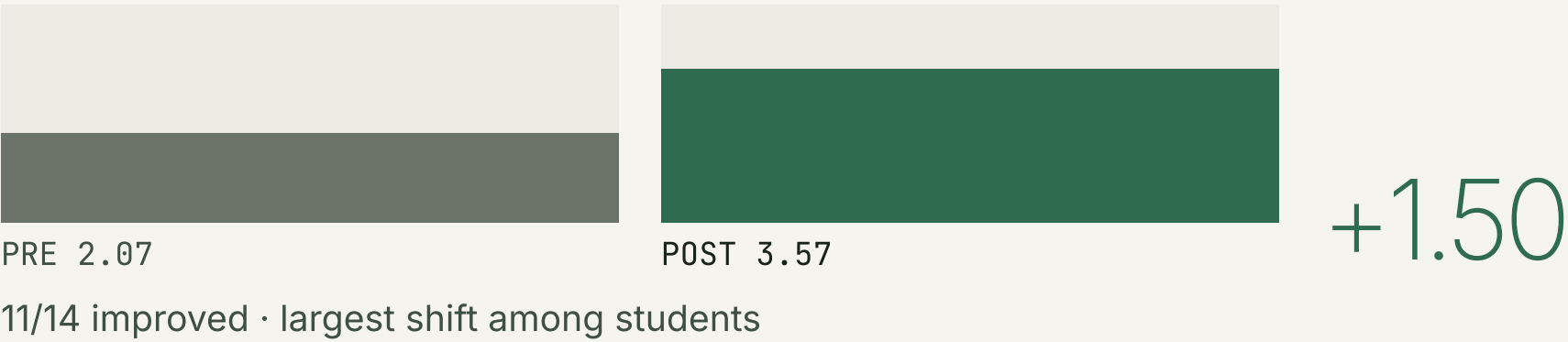
DATA ANALYSIS PLAN

Sample stratification by identity · data cleaning & duplicate checks · pre/post survey comparison · subgroup patterns · route/outcome coding · descriptive statistics first

DESCRIPTIVE RESULTS • SURVEY + AI ANALYSIS

Four early signals from completed surveys

PRE / POST UNDERSTANDING • N=14



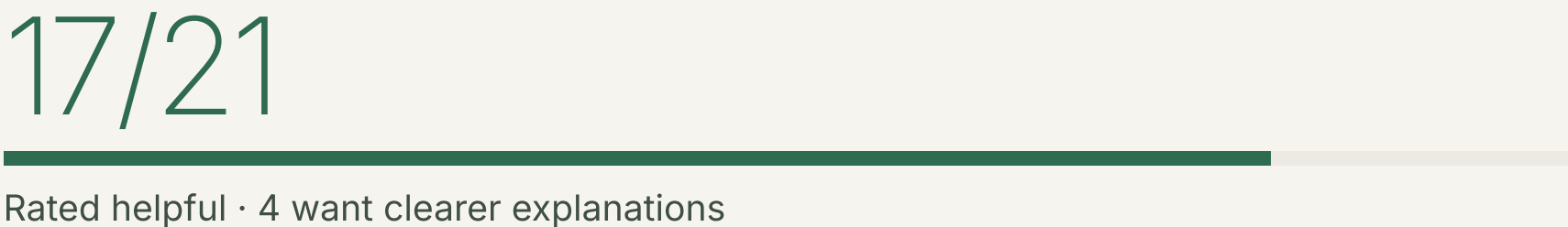
FARMER ROUTE CHOICE • TURN 3 • N=12



RECOMMENDATION • N=21



AI DEBRIEF • N=21



VALID SESSIONS

28
A+B pool

COMPLETED SURVEY

21
primary sample

TAKEAWAY

Awareness gain + positive reception; farmer-path claims remain formative.

PRIOR WORK VS. THIS STUDY • PRACTICAL IMPLICATIONS

From regional evidence
to household-level decisions.

DIFFERENCE FROM PRIOR WORK

Earlier studies mainly prove pollution, carbon, and affordability at regional or policy level.

This sandbox links household inputs with turn-by-turn route choices and terminal outcomes, so the question becomes which coal-to-X path fits this household, not only whether clean heating is good.



PRACTICAL IMPLICATIONS

The sandbox turns abstract policy trade-offs into decisions that households can try before they commit.

Future farmers can preview the coal-to-X path that best fits their own household before switching. Village governments can then aggregate these household inputs and choices to select the most appropriate plan for the whole village.

Sources cited in this study.

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Q & A

Thank you
Questions
welcome

Village Clean-Heating Transition Sandbox

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DEMO

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INSTRUCTOR

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COURSE

SCP KU